RAJALAKSHMI ENGINEERING COLLEGE

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R2

Started on	Sunday, 30 August 2026, 8:19 PM
State	Finished
Completed on	Sunday, 30 August 2026, 8:20 PM
Time taken	38 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Problem Statement:

Given a sorted array and a value x, the floor of x is the largest element in array smaller than or equal to x. Write divide and conquer algorithm to find floor of x.

Input Format

First Line Contains Integer n – Size of array

Next n lines Contains n numbers – Elements of an array

Last Line Contains Integer x – Value for x

Output Format

First Line Contains Integer – Floor value for x

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>
#include <stdlib.h>

// Divide and Conquer function to find the floor of x
int findFloor(int arr[], int low, int high, int x) {
    // If the search space is exhausted and x is smaller than the first element
    if (low > high) {
        return -1;
    }

    // If x is greater than or equal to the last element, the last element is the floo
    if (x >= arr[high]) {
        return arr[high];
    }

    int mid = low + (high - low) / 2;

    // If mid element itself is x, it is the floor
    if (arr[mid] == x) {
        return arr[mid];
    }

    // If x lies between mid-1 and mid, then mid-1 is the floor
    if (mid > 0 && arr[mid - 1] <= x && x < arr[mid]) {
        return arr[mid - 1];
    }

    // If x is smaller than mid element, search the left half
    if (x < arr[mid]) {
        return findFloor(arr, low, mid - 1, x);
    }

    // If x is greater than mid element, search the right half
    return findFloor(arr, mid + 1, high, x);
}

int main() {
    int n;

    // Read the size of the array
    if (scanf("%d", &n) != 1 || n <= 0) {
        return 1;
    }

    int* arr = (int*)malloc(n * sizeof(int));
    if (arr == NULL) {
        return 1;
    }

    // Read the next n lines for array elements
    for (int i = 0; i < n; i++) {
```

	Input	Expected	Got	
✔	6 1 2 8 10 12 19 5	2	2	✔
✔	5 10 22 85 108 129 100	85	85	✔
✔	7 3 5 7 9 11 13 15 10	9	9	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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